Lista 1 – MLG

1a) $f(y|\mu) = \frac{4(y+1)\mu^y}{(\mu+2)^{y+2}} = \exp\{\log 4(y+1) + y\log\mu - (y+2)\log(\mu+2)\} =$

$= \exp\{1[y(\log\mu/(\mu+2)) - 2\log(\mu+2)] + \log 4(y+1)\}$

$\rightsquigarrow \phi = 1,\ \theta = \log\mu/(\mu+2),\ b(\theta) = -2\log(1-e^\theta)/2,\ c(\phi, y) = \log 4(y+1)$,,

b) $E(Y) = b'(\theta) = -2 \cdot \frac{1}{1-e^\theta} \cdot -e^\theta = 2 \cdot \frac{1}{1-(\mu/\mu+2)} \cdot \mu/\mu+2 = \mu$,,

$Var(Y) = 2[(1-e^\theta)e^\theta - e^\theta \cdot (-e^\theta)] \cdot 1/(1-e^\theta)^2 = 2 \cdot e^\theta/(1-e^\theta)^2 = \frac{2\mu/\mu+2}{(1-\mu/\mu+2)^2} =$

$= \frac{\mu(\mu+2)}{2}$,,

c) $D(\underset{\sim}{y}, \hat{\mu}) = 2 \cdot \mathcal{L}[y_i(\log(y_i/y_i+2) - \log(\hat{\mu_i}/\hat{\mu_i}+2)) + (-2\log(y_i+2)) - (-2\log(\hat{\mu_i}+2))] =$

$= \mathcal{L}[2y_i \cdot \log(y_i(\hat{\mu_i}+2)/\hat{\mu_i}(y_i+2)) - 4\log(y_i+2/\hat{\mu_i}+2)]$,,

2a) $f(y|p) = p^y(1-p)^{1-y} = \exp\{y\log p + (1-y)\log 1-p\} = \exp\{1[y(\log(p/1-p)) + \log 1-p]\}$

$\rightsquigarrow \phi = 1,\ \theta = \log(p/1-p),\ b(\theta) = \log(1+e^\theta),\ c(\phi, y) = 0$,,

b) $E(Y) = b'(\theta) = \frac{e^\theta}{1+e^\theta} = \frac{p/1-p}{1+p/1-p} = p$,,

$Var(Y) = \phi^{-1} \cdot b''(\theta) = \frac{e^\theta(1+e^\theta) - e^{2\theta}}{(1+e^\theta)^2} = \frac{e^\theta}{(1+e^\theta)^2} = \frac{p/1-p}{(1+p/1-p)^2} = p(1-p)$,,

c) $D(\underset{\sim}{y}, \hat{p}) = 2 \cdot \mathcal{L}[y_i(\log(y_i/1-y_i) - \log(\hat{p}/1-\hat{p})) + \log 1-y_i - \log 1-\mu_i] =$

$= 2 \cdot \mathcal{L}[y_i(\log(y_i(1-\hat{p})/\hat{p}(1-y_i)) + \log(1-y_i/1-\hat{p})]$,,

3-) $f(y|p) = p^y/\{-y\log(1-p)\} = \exp\{y\log p - \log(-y) - \log(\log(1-p))\}$

$\rightsquigarrow \phi = 1,\ \theta = \log p,\ b(\theta) = \log(-\log(1-e^\theta)),\ c(\phi, y) = -\log(-y)$,,

$E(Y) = b'(\theta) = \frac{e^\theta}{-\log(1-e^\theta)(1-e^\theta)} = \frac{p}{-\log(1-p)(1-p)}$,,

$V(y) = b''(\theta) = \frac{e^\theta \log(1-e^\theta)(1-e^\theta) - e^\theta\left(\frac{-e^\theta(1-e^\theta)}{(1-e^\theta)} + -e^\theta \cdot \log(1-e^\theta)\right)}{\log(1-e^\theta)^2(1-e^\theta)^2}$

$= \frac{-p\log(1-p)(1-p) + p^2 + p^2\log(1-p)}{\log(1-p)^2(1-p)^2} = \frac{-p(\log(1-p)+p)}{\log(1-p)^2(1-p)^2}$,,

$\mathrm{Log}(1-\hat{\mu})/(1-\hat{\mu})$ $\mathrm{Log}(1-\gamma_i)/(1-\gamma_i)$

$D(\gamma;\hat{\mu}) = 2\cdot\sum\{\gamma_i(\mathrm{Log}\,\gamma_i - \mathrm{Log}\,\hat{\mu}) - \mathrm{Log}(-\mathrm{Log}(1-\gamma_i)) + \mathrm{Log}(-\mathrm{Log}(1-\hat{\mu}))\} =$

$= 2\cdot\sum[\gamma_i(\mathrm{Log}\,\gamma_i/\hat{\mu}) + \mathrm{Log}[\mathrm{Log}(1-\hat{\mu})/\mathrm{Log}(1-\gamma_i)]$ //

4a) $\eta_i = g(\mu_i) = -\mathrm{Log}(-\mathrm{Log}(\mu_i)) \Rightarrow -\mathrm{Log}\,\mu_i = e^{-\eta_i} \Rightarrow \mu_i = e^{-e^{-\eta_i}}$ //

b) $\hat{\mu}_i = e^{-e^{-x_i\hat{\beta}}} = e^{-e^{-x^T\hat{\beta}}}$ ↝ sabemos que $e^{-e^{-x^T\hat{\beta}}} > 0$, temos que verificar se é < 1

↝ sabemos que $e^x < 1$ se $x < 0 \Rightarrow$ precisamos que $-e^{-x^T\beta} < 0$, o que é verdadeiro

$\Rightarrow 0 < e^{-e^{-x^T\hat{\beta}}} < 1 \Rightarrow 0 < \hat{\mu_i} < 1$ //

5a) Como visto em aula, para a gama com $\mu = d\cdot\beta$ e $\lambda = d$, temos que

$\theta_i = -1/\mu_i$, $b(\theta_i) = -\mathrm{Log}(-\theta_i) = \mathrm{Log}(\mu_i)$

↝ por a função de ligação ser a canônica temos que $w_i = v_i = 1/\theta^2 = \mu_i^2$ //

↝ $g(\mu_i) = \eta_i = -1/\mu_i \Rightarrow \mu_i = -1/\eta_i$

$\Rightarrow z_i = \eta_i + G_{ii}(\gamma_i - \mu_i)$, $G_{ii} = \dfrac{\partial\eta_i}{\partial\mu_i} = \dfrac{1}{\mu_i^2}$

$\Rightarrow z_i = \dfrac{-1}{\mu_i} + \dfrac{\gamma_i - \mu_i}{\mu_i^2} = \dfrac{-\mu_i + \gamma_i - \mu_i}{\mu_i^2} = \dfrac{\gamma_i - 2\mu_i}{\mu_i^2}$ //

b) Sob função de ligação logarítmica temos que $g(\mu_i) = \mathrm{Log}\,\mu_i = \eta_i \Rightarrow \mu_i = e^{\eta_i}$

$\Rightarrow w_i = \dfrac{1}{v_i}\cdot\left(\dfrac{\partial\mu_i}{\partial\eta_i}\right)^2 = \dfrac{1}{\mu_i^2}\cdot(e^{\eta_i})^2 = \dfrac{1}{(e^{\eta_i})^2}\cdot(e^{\eta_i})^2 = 1 \Rightarrow W$ é a identidade //

↝ $z_i = \eta_i + G_{ii}(\gamma_i - \mu_i)$, $G_{ii} = \dfrac{1}{\mu_i}$

$\Rightarrow z_i = \mathrm{Log}\,\mu_i + \dfrac{\gamma_i - \mu_i}{\mu_i}$ //

6-) $f(\gamma;\mu,\sigma^2) = \dfrac{1}{\sqrt{2\pi\sigma^2}}\cdot\exp\left\{\dfrac{-1}{2\sigma^2}\cdot(\gamma-\mu)^2\right\} = \exp\left\{-\mathrm{Log}\sqrt{2\pi\sigma^2} - \dfrac{(\gamma^2 - 2\gamma\mu + \mu^2)}{2\sigma^2}\right\} =$

$= \exp\left\{\dfrac{1}{\sigma^2}\left(\gamma\cdot\mu - \dfrac{\mu^2}{2}\right) - \left(\dfrac{\gamma}{2\sigma^2} + \dfrac{\mathrm{Log}\,2\pi\sigma^2}{2}\right)\right\}$

↝ $\phi = \dfrac{1}{\sigma^2}$, $\theta = \mu$, $b(\theta) = -\dfrac{\mu^2}{2} = -\dfrac{\theta^2}{2}$, $c(\phi,\gamma) = -\left(\dfrac{\gamma^2}{2\sigma^2} + \dfrac{\mathrm{Log}\,2\pi\sigma^2}{2}\right)$ //

↝ Temos que $D^*(\underset{\sim}{\gamma},\underset{\sim}{\hat{\mu}}) = \phi^{-1}D(\gamma,\hat{\mu}) = 2\cdot\sum[\gamma_i(\gamma_i - \hat{\mu_i}) + \hat{\mu_i}^2/2 - \gamma_i^2/2] =$

$= 2 \cdot \sum [y_i^2/2 - y_i \hat{\mu}_i + \hat{\mu}_i^2/2] = \sum (y_i - \hat{\mu}_i)^2$

$\rightsquigarrow U_\phi = \sum [y_i \theta_i - b(\theta_i)] + \sum \frac{\partial c(y_i, \phi)}{\partial \phi} = \sum [y_i \cdot \hat{\mu}_i + \hat{\mu}^2/2] + \sum -(y_i^2/2 + (-1/2\phi)] =$

$= \sum [y_i \hat{\mu}_i - \hat{\mu}_i^2/2 - y_i^2/2] + n/2\hat{\phi} = 0 \Rightarrow \frac{n}{2\hat{\phi}} = \frac{1}{2} \cdot \sum (y_i - \hat{\mu}_i)^2$

$\Rightarrow \hat{\phi} = \frac{n}{\sum (y_i - \mu_i)^2} = \frac{n}{D^*(\underset{\sim}{y}, \underset{\sim}{\hat{\mu}})}$ //

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